

Development of Antenna-Coupled Detectors and a Small Aperture Telescope for Cosmic
Microwave Background Observations

by

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Abstract

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Cosmic microwave background (CMB) experiments have paved the way towards a greater understanding of early universe physics and building a standard model of cosmology. In this dissertation, we report on the development, integration, testing, and deployment of the ultra-high-frequency small aperture telescope for the Simons Observatory (SO). We also discuss development and optimization of detector arrays for CMB experiments, and describe the SO low-frequency wafers in detail. This work also directly impacts developments for CMB-S4 and *LiteBIRD*. We also discuss the ongoing calibration and scan strategy efforts in SO as the experiment moves towards initial science observations.

To my father, who is my best friend and has stood by me in my worst and best times,
always there to catch me if I stumble. And to my mother, who has taught me simplicity,
discipline, and the power of staying true to myself.

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